Randomized and low-rank approximation

The Goreinov–Tyrtyshnikov–Zamarashkin conjecture on square-submatrix inverse norms

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Improving the classical bound is challenging and would sharpen stability guarantees for skeleton/CUR approximation.

Problem statement

For integers $1 \leq r < n$, let $U \in \mathbb{R}^{n\times r}$ satisfy $U^TU = I_r$. Write $U_I = U(I,:)$ for $I \subseteq \{1,\dots,n\}$ with $|I| = r$, and let $\|\cdot\|_2$ denote the spectral norm.

Conjecture (Goreinov–Tyrtyshnikov–Zamarashkin [1, Eq. (2.6)]). Every such U admits a nonsingular square submatrix U_I satisfying

$$\|U_I^{-1}\|_2 \leq \sqrt{n}.$$

In the notation of [1], this is $t(r,n) \leq \sqrt{n}$, where

$$t(r,n) = \max_{\substack{U\in\mathbb{R}^{n\times r}\\ U^TU=I_r}} \quad \min_{\substack{I\subseteq\{1,\dots,n\}\\ |I|=r,\ \det(U_I)\neq 0}} \|U_I^{-1}\|_2.$$

The minimum is over nonsingular submatrices; at least one exists because $\operatorname{rank}(U) = r$.

Known bounds and cases

The bound

$$t(r,n) \leq \sqrt{r(n-r)+1}$$

is well known and follows from various arguments: maximal volume [1], Bischof–Stewart QR (BSQR) [2, 3], or volume sampling [4, 5] just to name a few. Also, $t(r,n) \geq \sqrt{n}$ whenever $r+1$ divides n [1, Lem. 2.2], so the conjectured constant is sharp.

The real case $r = 1$ is elementary. Nesterenko [6] proved the real case $n = 4$, $r = 2$; Sengupta–Pautov [7] subsequently proved $r = 2$ for all n. Orthogonal completion gives $t(r,n) = t(n-r,n)$, leaving $3 \leq r \leq n-3$ as the general open range. These inverse norms control pseudoskeleton approximation errors [1, Thms. 3.1–3.2].

Proposed complex extension

Define $t_{\mathbb{C}}(r,n)$ analogously using $U \in \mathbb{C}^{n\times r}$ with $U^*U = I_r$, where $*$ denotes conjugate transpose. The minimization again runs only over nonsingular U_I. The classical bound $\sqrt{r(n-r)+1}$ also holds over $\mathbb{C}$ [3].

Conjecture (proposed by the contributor). There is an absolute constant α, independent of r and n, such that

$$t_{\mathbb{C}}(r,n) \leq \alpha\sqrt{n}, \qquad 1 \leq r < n.$$

The dependence on r, n and the best possible α remain to be determined. The constant $\alpha = 1$ fails: $t_{\mathbb{C}}(2,4) = \sqrt{3+\sqrt{3}} > 2$ [8, 9]. For two columns, Nesterenko [9, Prop. 1] proves

$$t_{\mathbb{C}}(2,n) \leq c_2\sqrt{n}, \qquad c_2 = \left(2 - \frac{2}{\sqrt{3}}\right)^{-1/2} \approx 1.08766,$$

with equality whenever $4 \mid n$, so any universal α must satisfy $\alpha \geq c_2$.

References

1. S. A. Goreinov, E. E. Tyrtyshnikov, and N. L. Zamarashkin, *A theory of pseudoskeleton approximations*, Linear Algebra and its Applications **261** (1997), 1–21. Published paper.
2. A. Damle, *Computing Strong Rank-Revealing Factorizations for Matrices with Orthonormal Rows*, arXiv: 2607.13532v1 (2026). Preprint.
3. A. I. Osinsky, *Close to optimal column approximation using a single SVD*, Linear Algebra and its Applications **725** (2025), 359–377. Published paper.
4. H. Avron and C. Boutsidis, *Faster subset selection for matrices and applications*, SIAM Journal on Matrix Analysis and Applications **34** (2013), 1464–1499. Published paper.
5. A. Cortinovis and D. Kressner, *Adaptive Randomized Pivoting for Column Subset Selection, DEIM, and Low-Rank Approximation*, SIAM Journal on Matrix Analysis and Applications **47** (2026), 25–47. Published paper.
6. Y. Nesterenko, *Submatrices with the best-bounded inverses: revisiting the hypothesis*, arXiv: 2303.07492 (2023; revised 2024). Preprint.
7. R. Sengupta and M. Pautov, *On the submatrices with the best-bounded inverses*, arXiv: 2604.05944v5 (2026). Preprint.
8. Y. Nesterenko, *Submatrices with the best-bounded inverses: Studying $\mathbb{R}^{n\times 2}$ and $\mathbb{C}^{n\times 2}$*, arXiv: 2408.16631v1 (2024). Preprint.
9. Y. Nesterenko, *Submatrices with the best-bounded inverses: an asymptotically tight upper bound for $\mathbb{C}^{n\times 2}$*, arXiv: 2604.24087v1 (2026). Preprint.

Status check — 2026-09-11

Checked [1]–[9] and searched for later resolutions; no general real-case resolution or resolution of the proposed complex extension was found. The real two-column result is a preprint and supports the partially resolved status.